

Principles of Biology 413

Fall 2016

Instructor: Dr. Patricia A. Halpin Patricia.Halpin@unh.edu 641-4140

Office hours: Monday 1-2 & Wednesdays 4-5pm or by appointment. Rm555

Textbook: Campbell Biology in Focus (2014) by Urry, et al., 1st edition 0321813804 hardcover or 0321896882 looseleaf. You can also get the second edition.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted. Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Lab Manual: none

A lab coat is required for admittance to lab and may be purchased at the UNH bookstore <http://goo.gl/1sZett>

Lecture: Wednesdays 9-11:50 am P301.

As we meet only once per week it is a week's worth of material presented in one class period. To achieve success in this course it is necessary to read the material before class and review the material daily. I highly encourage you to go to study group weekly. Ask questions if you do not understand. Check your UNH email account often. This is the one I will use to contact you outside of class. You can also use Facebook messages to contact me.

UNHM Biology & Biotechnology Facebook Page

At the suggestion of students this page was created for students and alumni to stay connected to what is going on on-campus and opportunities for you to enhance your career and build your

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professional network. This is professional use of social media and all are expected to become a member. Join now.

Quizzes: The weekly quizzes are located in Canvas and should be taken within the week after lecture. For example the lecture is Wednesday the quiz is available Wednesday at 1 pm and ends the following Tuesday at midnight. Always take the quiz as they cannot be made up. The two lowest grades of quizzes taken will be dropped. If you do not take a quiz the grade (zero) will count. There are no makeups of quizzes.

Exams: All exams must be taken when scheduled. Make-up exams will be permitted only for excused absences (e.g., confirmed illness with doctor's note) and advance notice of absence. Each exam only covers material since the previous exam; i.e., the second and third exams are not cumulative. Your course grade will be determined based on the percentage of points earned out of the possible 500. 90% and above will earn A- or better; 80% and above will earn B- or better; 70% and above will earn C- or better; 60% and above will earn D- or better. This is college there is no extra credit.

Attendance: Attendance in both lecture and laboratory is expected. Although I usually do not take attendance during the lecture, you are responsible for all information and material presented during the class. Since the laboratory is a hands-on experience, attendance is taken. Excessive absences (two or more) from the laboratory will result in a reduction in your grade.

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<u>Dates</u>	<u>Lecture # & Topic</u>	<u>Chapter #</u>
August 31	1. Introduction; What is Biology? And scientific method	1
September 7	2. . The Chemical Context of Life	2
September 14	3. The Molecular Diversity of Life	3
September 21	4. A Tour of the Cell	4
September 28	5. Membrane Transport and Cell Signaling	5
October 5	<i>Exam 1</i>	
October 12	6. The Cell Cycle and Meiosis	9-10
October 19	7. Inheritance, Genes and Chromosomes	11-12
October 26	8. The Molecular Basis of Inheritance	13
November 2	9. Gene Expression: From Gene to Protein	14
November 9	<i>Exam 2</i>	
November 16	10. Gene Regulation, Stem Cells, and Viruses	15-17
November 23	No Class Thanksgiving Break	
November 30 6	11. Genomes, Recombinant DNA and Biotechnology	18,
	11. Introduction to Metabolism	
December 7	12. Cellular Respiration, Photosynthesis	7, 8
December 14	<i>Final exam</i>	

Grading: 3 lecture exams @ 100 points each..... 300

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Best 10 of 12 weekly quizzes **taken** @ 10 points each.....100

Laboratory 125

Total Points Available..... 525

BIOL 413 Laboratory

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Contact Information:

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Rm 550B

Feel free to contact me at any time. Email works best, I will attempt to respond to you as soon as possible.

Lab Schedule:

Module 1: Scientific Method

Aug 31, Sep 1, 2 Safety, Introduction to Brine Shrimp and Scientific Method

Sep 7, 8, 9 Experimental Design

Sep 14, 15, 16 Run Experiment, Data Collection, Statistical Analysis

Sep 21, 22, 23 Communicating in Science; Anatomy of a Journal Article, Conventions,
Presentations ****Quiz 1 brine shrimp**

Sep 28, 29, 30 Student Presentations

Module 2: Cells

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Oct 5, 6, 7 Mitosis, Meiosis and Cell Division

Module 3: Molecular Genetics

Oct 12, 13, 14 Crime Scene Evidence Collecting & Blood Typing

*****Quiz 2 Mitosis/Meiosis***

Oct 19, 20, 21 Crime Scene ELISA Test

*****Quiz 3 Blood Typing***

Oct 26, 27, 28 Crime Scene Restriction Fragment Analysis

***** Quiz 4 ELISA***

Nov 2, 3, 4
report Crime Scene Final Analysis & Communicating in Science – Forensic

Nov 9, 10, 11 **Veterans Day No Labs**

Nov 16, 17, 18 Student Presentations

***** Quiz 5 RFLP***

Nov 23, 24, 25 **Thanksgiving No Labs**

Module 4: Cellular Respiration and Metabolism

Nov 30, Dec 1, 2 Metabolism

Dec 7, 8, 9 **Lab Exam**

Grading:

Lab grade will be a total of 125pts made up of the following:

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Presentation/Lab Report from Module 1	20pts
Presentation/Lab Report from module 3	20pts
Lab Report from Module 2	10pts
Lab Report from module 4	10pts
Lab Quizzes	25pts
Lab Exam	20pts
Attendance	20pts
Total	125pts

Attendance:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Also we will be working as groups during this lab and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab.

Canvas:

All of the materials for this lab are found in your lab section of Canvas. While there may be times where you will be given hard copies of materials you will need for lab there will also be times where you will need to print out and bring the materials to lab. It is also a good way for you to review any videos or interactive programs that we may use in lab.

Electronic Devices:

You are welcome to use electronic devices (cell phones, tablets, laptops) in lab for “lab related” purposes. If you however need, or any other non-lab related use of these devices (i.e. to take or place a phone call) please be respectful of your lab mates and go outside the lab for such activities.

Lab Notebook:

You will need a notebook for lab to record your experimental progress and data in.

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Lab Manual:

There is no lab manual that needs to be purchased for this lab.

To do well in this lab:

- Attend all labs! If you miss a lab you will miss an experience that cannot be replicated by simply reading the lab.
- Be prepared when coming to lab. Most material is available in advance of the lab meeting time on the Blackboard site.
- Make notes and sketches during lab. Ask questions, ask questions, ask questions!
- Fill out lab reports completely, and answer questions with more than a word or two.
- Study for the practical exam *every week* by reviewing all handouts, lab reports, notes & sketches. If you wait till just before the practical it will be too much information.

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4170, jenessa.zurek@unh.edu or in person in the Student Services Suite, Room (#410H).